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List of Abbreviations

Abbreviation	Definition
AM	Alma Mater (refers to UNIBO – University of Bologna)
ANVUR	Agenzia Nazionale di Valutazione del Sistema Universitario e della Ricerca
BBS	Bologna Business School
CDTI	Centre for the Development of Industrial Technology (Spain)
CIR	Interdepartmental Centers for Industrial Research (UNIBO)
CNFPT	Centre National de la Fonction Publique Territoriale (France)
CNRS	Centre National de la Recherche Scientifique (France)
CoARA	Coalition for Advancing Research Assessment
COO	Coordinator (role in project)
DAE	Direction des Achats de l'État (France)
EC	European Commission
ECMWF	European Centre for Medium-Range Weather Forecasts
EFSA	European Food Safety Authority
EOSC	European Open Science Cloud
ERA	European Research Area
EIE	European Innovation Ecosystem
EIT	European Institute of Innovation and Technology
ES	Spain
ESIF	European Structural and Investment Funds
EU	European Union
FAIR	Findable, Accessible, Interoperable, Reusable (data principles)
FECYT	Fundación Española para la Ciencia y la Tecnología
HEIs	Higher Education Institutions
IAM	Intellectual Asset Management
ICT	Information and Communication Technology
IGRPE	Institut de Gestion Publique et de l'Economie (France)
IP	Intellectual Property
IPR	Intellectual Property Rights
ISCIII	Instituto de Salud Carlos III (Spain)
IT	Italy
ITAEOSC	Italian EOSC Tripartite Assembly
JPIs	Joint Programming Initiatives
KEI	Knowledge Exchange and Impact
KIC	Knowledge and Innovation Community
KTO	Knowledge Transfer Office
KT	Knowledge Transfer
KV	Knowledge Valorisation
LC	Large Company
LPR	Loi de Programmation de la Recherche (France)
MLE	Mutual Learning Exercise
MUR	Ministero dell'Università e della Ricerca (Italy)
NGO	Non-Governmental Organisation
NRRP (or PNRR)	National Recovery and Resilience Plan
OECD	Organisation for Economic Co-operation and Development
PHI	Public Health Institution
PIA	Programme d'Investissements d'Avenir
PNR	Programma Nazionale per la Ricerca
POC	Proof of Concept
PROs	Public Research Organisations
PRUAB	Fundació Parc de Recerca UAB
R&I	Research and Innovation
R2B	Research to Business (event)
RRF	Recovery and Resilience Facility
RIS3	Research and Innovation Strategies for Smart Specialisation
RTD	Research and Technological Development
RTO	Research and Technology Organisation
SBA	Sistema Bibliotecario di Ateneo (UNIBO Library System)
SE	Sweden
SEN	Sensitive (dissemination level)
SIP	Strategic Innovation Programme
SME	Small and Medium-sized Enterprise
SNRI	Stratégie Nationale de Recherche et d'Innovation (France)
SSH	Social Sciences and Humanities
S3	Smart Specialisation Strategy
TTO	Technology Transfer Office
UAB	Universitat Autònoma de Barcelona
UH	University of Haifa
UL	Université de Lorraine
UNIBO	Alma Mater Studiorum – Università di Bologna
USF	University Seed Fund
WP	Work Package

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Executive Summary

Knowledge valorisation, as defined by the European Commission, is the process of transforming research and innovation results into sustainable economic, social, and environmental value. This concept signals a move away from traditional linear models of knowledge transfer toward a more systemic and contextual approach—where the reuse of knowledge, cross-sector collaboration, and impact-driven strategies are central. In the context of the KALEIDOS project, we adopt and build on this shift by conducting a structured review of the European knowledge valorisation landscape.

We mapped key EU-level initiatives and analysed how they cascade into national strategies. In doing so, we also reflected on one of the recurring challenges that many institutions encounter: how to create alignment between open science practices and knowledge transfer mechanisms. Rather than treating open and closed science as separate or opposing domains, we explored how they can interact and reinforce each other in practice. These “inter-actions”¹—for instance, between early sharing of results and strategic protection of assets—can support organisations in navigating decisions about when and how to share, protect, or reuse knowledge. This perspective informed the development of our Knowledge Valorisation Readiness Self-Assessment Tool. The tool supports organisations—particularly universities—in assessing their maturity across key valorisation channels and enabling conditions, offering a structured framework to analyse the interplay of actors, infrastructures, and knowledge flows within the Quadruple Helix model, helping institutions align their strategies with EU-level goals and improve their readiness over time.

Rather than treating intellectual property management as a constraint to openness, we frame it as part of a broader institutional strategy that balances protection with responsible sharing. From this perspective, bridging open and closed science models requires integrated approaches, organisational interoperability, and context-sensitive tools. Building on the insights generated throughout this work, we conclude the deliverable by outlining a set of policy guidelines. These are intended to inform and support institutions and policymakers and to serve as a foundation for formulating more structured and actionable policy recommendations that advance inclusive, trust-based innovation ecosystems aligned with EU priorities.

¹ EU Directorate-General for Research and Innovation, Prosperity Directorate. (2022). Open Science and Intellectual Property Rights: How can they better interact? State of the art and reflections (Report of Study).

1. Introduction

In the current European and global research landscape, there is an urgent need to explore interactions between Open Science practices with the management of Intellectual Property Rights (IPRs) and knowledge transfer mechanisms². Despite increasing political commitment to knowledge valorisation, many institutions and ecosystems still face structural and cultural barriers that prevent the full circulation of research-based knowledge. These include fragmented governance, underdeveloped infrastructures, rigid incentive systems, and a lack of coordination across sectors and levels of action³. At the same time, knowledge valorisation is often interpreted narrowly—as commercial exploitation or intellectual property management—leaving out other relevant forms such as citizen engagement, policy uptake, social innovation, and standardisation⁴. This narrow view can marginalise societal actors and limit the transformative potential of knowledge within complex systems⁵. Hence, several tensions emerge when attempting to operationalise knowledge valorisation in the quadruple helix. Policies may promote impact, but institutions lack the mechanisms and tools to act on them effectively. Valorisation efforts are often confined to single organisations, while real impact depends on cross-sector collaboration. Moreover, scientific outputs may be valorised, but experiential or community knowledge often remains invisible or under-utilized. In addition, actors within the quadruple helix face distinct needs and asymmetries in resources, mandates, and legitimacy. Higher education institutions (HEIs) are often positioned as central drivers of knowledge valorisation but face internal constraints related to reward systems, siloed expertise, or limited engagement infrastructures. Hasche et al. (2020) highlight the need for further research on the quadruple helix model from a micro-perspective, focusing on dynamic relationships, synergies, collaborations, coordinated environments, and value-creation activities. In this context, the challenge is not merely to promote knowledge valorisation but to create enabling conditions that support its systemic, inclusive, and mission-oriented implementation — while aligning with open science principles and national/EU policy priorities.

Starting in 2021, the European Commission delivered a series of principles and recommendations aimed at embracing a new vision for “Knowledge Valorisation” (hereby referred to as KV) for the following years. The expression of KV has a precise connotation, which can be put at the intersection of IP protection actions and open science. Traditional knowledge transfer, as outlined in the EC 2008 Recommendation⁶, was rightfully centred on technology and its commercialization. While not opposing the traditional focus on technology transfer and IP commercialisation—as outlined in the 2008 EC Recommendation—the concept of knowledge valorisation expands the scope to include all types of knowledge (scientific, technical, social, and cultural) and all actors within the R&I ecosystem. It moves beyond the valorisation of technological assets alone to embrace a more inclusive, mission-oriented approach that acknowledges the contribution of diverse knowledge forms and practices to societal and environmental challenges.

This policy brief advances the knowledge valorisation agenda by introducing a structured policy tool designed for Higher Education Institutions (HEIs) to assess their maturity in knowledge valorisation. At the heart of this contribution is a self-assessment maturity model, developed and validated through an in-depth case study of the University of Bologna. As the “case zero,” this pilot enables contextualised diagnostics and paves the way for broader deployment across all consortium countries. Grounded in a systemic understanding of research and innovation (R&I) ecosystems, the tool acknowledges intellectual assets as dynamic enablers of transformation—activated through aligned governance frameworks, targeted incentives, fit-for-purpose infrastructures, and a culture of cross-sectoral collaboration. By operationalising these principles, the tool not only maps existing capacities but also guides strategic development at institutional, national, and European levels.

✳ How to read this document/deliverable

This report provides both a conceptual foundation and practical guidance for operationalising knowledge valorisation across higher education institutions and national innovation ecosystems. It reflects the strategic shift promoted by the European Commission—from linear knowledge transfer to a multidimensional, mission-oriented, and inclusive vision of knowledge use. The document combines comparative policy analysis, institutional diagnostics, and actionable recommendations to support capacity building and alignment with the EU Guiding Principles for Knowledge Valorisation.

To facilitate navigation, the report is structured as follows:

² European Commission: Directorate-General for Research and Innovation, Open science and intellectual property rights –How can they better interact? –State of the art and reflections –Executive summary, Publications Office of the European Union, 2022, <https://data.europa.eu/doi/10.2777/3473054>

³ European Commission: Directorate-General for Research and Innovation and Cruz, K., Policy and legal context, governance and funding – MLE – Focus on skills, intersectoral cooperation and incentive systems – Topic 2a thematic report, Cruz, K.(editor), Publications Office of the European Union, 2024, <https://data.europa.eu/doi/10.2777/231623>

⁴ In the EU R&I context, standardisation is defined as a distinct knowledge and technology transfer channel that picks up on research outputs such as patents or publications and ensures their wider uptake by codifying knowledge into consensus-based specifications. Unlike patents, which indicate invention, standards guarantee interoperability, compatibility, and minimum quality or safety levels. Through this codification, standardisation facilitates the dissemination, usability, and application of R&I results across sectors and stakeholders. For more, please refer to European Commission: Directorate-General for Research and Innovation, ECORYS, EFIS, FH KREMS University of Applied Sciences Austria, ime, Radauer, A., Tardos, G., Baronowski, S., Yeghyan, M., Cowey, L., Boski, I., Schäfer, S. D., Teufel, B.Angelis, J., Scoping study for supporting the development of a code of practice for researchers on standardisation – Final report, Tardos, G.(editor), Publications Office of the European Union, 2022, <https://data.europa.eu/doi/10.2777/567608>

⁵ European Commission: Directorate-General for Research and Innovation, Kosova, H. and Vanrie, P., Mutual learning exercise on knowledge valorisation –Focus on skills, intersectoral cooperation and incentive systems –Final report, Kosova, H.(editor) and Vanrie, P.(editor), Publications Office of the European Union, 2024, <https://data.europa.eu/doi/10.2777/146102>

⁶ Commission Recommendation of 10 April 2008 on the management of intellectual property in knowledge transfer activities and Code of Practice for universities and other public research organisations. ELI: <http://data.europa.eu/eli/reco/2008/416/oj>

⇒ **Chapter 2 | THE EU CONTEXT FOR KV**

Chapter 2 presents a comparative mapping of national knowledge valorisation ecosystems across Italy, Spain, France, and Sweden. Instead of benchmarking, it adopts a systemic lens to highlight how different countries mobilise knowledge for societal value, examining governance structures, institutional arrangements, and alignment with EU frameworks.

⇒ **Chapter 3 | MAPPING KNOWLEDGE VALORISATION IN EUROPEAN AND NATIONAL ECOSYSTEMS**

Chapter 3 examines the main barriers and enablers to knowledge valorisation in the KALEIDOS consortium, with a focus on the interplay between intellectual property management and open science. It explores how these dynamics affect institutional practices and proposes ways to navigate their tensions through integrated strategies.

⇒ **Chapter 4 | KNOWLEDGE VALORISATION READINESS SELF-ASSESSMENT TOOL**

Chapter 4 introduces the Knowledge Valorisation Readiness Self-Assessment Tool developed within the KALEIDOS project. The tool supports higher education institutions and other actors in assessing their maturity across the six valorisation channels and three enabling dimensions—environmental, organisational, and individual.

⇒ **Chapter 5 | POLICY GUIDELINES**

Chapter 5 outlines ten policy guidelines for operationalising knowledge valorisation at institutional, national, and European levels. These guidelines are grounded in the EU and national initiatives, therefore, in the evidence, analyses, and practical insights generated throughout the mapping described in the previous chapters. Aligned with existing EU frameworks, they aim to support better actor alignment, reform incentive structures, and foster the development of inclusive and collaborative knowledge ecosystems.

⇒ **APPENDIX I & II**

Appendices include the full version of the self-assessment tool (Annex I), with diagnostic questions for internal use, and the University of Bologna case study (Annex II), which illustrates how the tool can be applied in practice to support strategic reflection and capacity building.

2. THE EU CONTEXT FOR KV

✳ The role of KTOs

Knowledge has become increasingly important in social and economic spheres, and universities are now seen as having a greater contribution to innovation. Historically, the core mission of universities has been the creation of knowledge through education and research activities (Gibbons et al., 1994), while the contemporary EU scenario leads universities to move beyond their historical isolation, often referred to as the "ivory tower" mentality (E3M project, 2011). This metaphor describes a detachment from practical concerns and a focus solely on internal intellectual pursuits. As a result, universities are now expected to play a more direct role in societal and economic development and are increasingly seen as active contributors to innovation. In response, this "new" role for universities emphasizes engagement with industry and society, transforming them into active participants in economic and cultural development.

According to "The Evolution of University-Based Knowledge Transfer Structures" report by the European University Association⁷, universities have historically struggled to connect their research with the outside world. To address this issue, universities created internal Knowledge Transfer Offices (KTOs) to assist researchers in navigating early-stage commercialisation. These units focus on pinpointing valuable research outputs, securing intellectual property rights, facilitating contractual arrangements, and laying the groundwork for collaborations with external actors. Originally, these offices played a key role in externally oriented communication and opportunity seeking, focusing on enabling connections with industry, supporting spin-offs, and handling sensitive legal and administrative procedures. As collaboration with industry intensified, however, their focus shifted toward providing more standardised, routine support for researchers within the university⁸. Moreover, activities closely tied to the market—such as partner matchmaking, business development, and investment attraction—are increasingly managed through cross-disciplinary managers or external actors, such as intermediaries positioned closer to the industry. These include university-affiliated incubators, joint laboratories, regional consortia, and science parks. Their emergence marks the transition toward what are known as European Innovation Ecosystems (EIE): interconnected networks that extend beyond the university walls, designed to promote the commercialisation of research in specialised domains and enhance the university's societal role "by drawing on the existing strengths of national, regional and local ecosystems"⁹. If the ability of universities to build effective innovation ecosystems depends on several factors, such as the maturity of the territorial innovation environment, the institution's investment capacity, and the alignment between its research culture and valorisation goals, studies show that universities with clear missions linking research to regional development are more successful in anchoring these ecosystems and attracting external actors¹⁰.

Moreover, a key insight from recent comparative studies is that not all universities engage in knowledge transfer in the same way or for the same reasons. Institutional characteristics – such as disciplinary focus (generalist vs. specialist) and prestige (high vs. low) – shape strategic choices in KT (Giuri et al., 2019). Prestigious and specialized universities tend to pursue income-generation strategies, focusing on patent licensing and the creation of spin-offs. Research-intensive institutions often follow a service-to-faculty model, supporting academic entrepreneurship and collaborative networks. In contrast, generalist and lower-prestige universities typically engage in local development strategies, emphasizing regional involvement and the promotion of local innovation ecosystems.

As for the role of national contexts in Europe, the governance of knowledge transfer reveals significant variation between centralized and decentralized policy mixes. A large-scale study of gap-funding instruments across 21 European countries highlights that the maturity of national TT infrastructures influences how support mechanisms—such as proof-of-concept (POC) programs and university seed funds (USFs)—are implemented (Munari et al., 2016). Countries with either nascent or very mature TT systems tend to adopt centralized instruments (e.g., national-level POCs). In contrast, decentralized models emerge in countries with intermediate levels of TT development, where regional governments or universities assume a more active role. This U-shaped relationship underscores the need for nuanced policy frameworks that are responsive to institutional maturity and context-specific needs.

✳ KV-led transformations

The Guiding Principles for Knowledge Valorisation (2022) emphasize that Open Science and valorisation should not be viewed as opposing logics but as complementary practices. At the centre of this shift is the recognition that openness is neither binary nor universal, but relational, strategic, and infrastructural. For this reason, the motto becomes a governance principle rather than a technical rule. It guides decisions on data sharing, licensing, collaborative models, and even staffing choices. It also reflects a shift from "technology momentum" to problem-oriented collaboration, where openness is not an end but a means to engage broader communities in solving complex challenges.

¹⁰ Grimaldi, R., Kenney, M. & Piccaluga, A. University technology transfer, regional specialization and local dynamics: lessons from Italy. *J Technol Transf* 46, 855–865 (2021). <https://doi.org/10.1007/s10961-020-09804-7>

As Ramachandran et al. (2021) argue, Open Science must be conceived not merely as the removal of barriers to data or publication, but as the creation of a collaborative culture enabled by technology, in which data, knowledge, and processes circulate efficiently and equitably across scientific and societal boundaries. Critically, this transformation entails a move from "open data" to "open science", where the latter encompasses not only access to outputs but a rethinking of the scientific process itself—including how problems are framed, who participates, and how impact is assessed (Ramachandran et al., 2021). This is particularly relevant for Knowledge Transfer Offices (KTOs), which must now operate at the interface of scientific norms, policy instruments, commercial strategies, and public interest. This is where ecosystem coordination becomes crucial. Ramachandran et al. (2021) emphasize that successful open science ecosystems require institutional investments in interoperability and collaborative infrastructures.

To balance openness with strategic direction, the EU now frames knowledge valorisation as a systemic function within the European Research Area (ERA). This shift moves away from isolated activities and narrow IP logics toward multi-actor ecosystems, co-creation platforms, and the broader management of intellectual assets. Valorisation today includes collaborative research, data commons, spin-offs, social innovation, citizen science, and is increasingly driven by Open Science principles. Over time, EU policy has evolved from fragmented national strategies to a more integrated, impact-driven approach. The current phase—anchored by the 2022 Guiding Principles—positions valorisation as a structural enabler of Europe's green, digital, and inclusive transitions. Key instruments such as Horizon Europe, the Recovery and Resilience Facility, EOSC, and Mutual Learning Exercises now support institutional maturity, interoperability, and multilevel coordination. These transitions are not rhetorical but structural, calling for an integrated approach that blends Open Science commitments with intellectual asset management, and that embeds valorisation capacities across governance levels, institutional cultures, and actor roles. To govern this complexity, the EU is moving toward what we call "interoperable ecosystems", which reflects the need for cross-actor, cross-level, and cross-infrastructure coordination. At the heart of this paradigm shift lies a fundamental tension in the expectations placed upon universities and public research institutions. On one hand, they are called to fully embrace the Open Science principles, ensuring FAIR data practices¹¹. On the other, they are expected to deliver measurable contributions to innovation, economic growth, and societal impact – objectives that often require the strategic protection and management of intellectual assets, particularly by IPRs, patents and technology transfer mechanisms¹².

✳ KV through systemic lens

As discussed, in the latest years, knowledge valorisation is increasingly conceptualised not as a set of isolated practices focused solely on commercialisation or intellectual property (IP) management, but as a multidimensional strategy that encompasses intellectual assets, co-created knowledge, intermediary functions, and systemic value generation across the quadruple helix of academia, industry, public authorities, and civil society (European Commission, 2022). The European Commission—through its Guiding Principles for Knowledge Valorisation—emphasises the need for a profound cultural, institutional, and systemic shift toward more inclusive and impact-driven research ecosystems.

This transformation has been underpinned by a series of strategic policy actions, including:

- *ERA Action 7*, which aims to foster a more coherent policy environment by providing common guidance and benchmarks for Member States. It supports the development and uptake of shared frameworks that enhance the translation of research results into social and economic value, in line with the broader ERA Policy Agenda (European Commission, 2022a).
- *The EU Codes of Practice*, which operationalise the Guiding Principles for KV by offering concrete methodologies and tools to support actors across the research and innovation landscape. It includes:
 - The Code of Practice on Intellectual Asset Management (IAM) expands the traditional focus on intellectual property by encouraging the strategic management of a wider array of assets—including data, software, know-how, and research tools—to enhance their value and reuse potential.
 - The Code on Co-Creation supports the development of collaborative frameworks between research-performing organisations, businesses, and civil society actors, fostering mutual benefit and relevance through participatory design.
 - The Code on Citizen Engagement emphasises the integration of public values and needs into the R&I process, aiming to increase legitimacy, trust, and the societal uptake of innovations.
 - The Code on Standardisation highlights the importance of aligning research outcomes with technical standards, thus facilitating interoperability, scalability, and faster deployment of results (European Commission, 2022b).
- *ERA Action 4*, which addresses the critical role of research careers in enabling knowledge valorisation. It focuses on improving working conditions, promoting intersectoral mobility, and increasing the attractiveness and

¹¹ The FAIR principles—Findable, Accessible, Interoperable, and Reusable—aim to enhance the discovery, integration, and reuse of research data by both humans and machines through persistent identifiers, rich metadata, and standardised protocols (Wilkinson et al., 2016).

¹² This particular point of view is presented by Alessandra Baccigotti in her article *Connecting the dots: IPRs, Knowledge Transfer, Innovation and Open Science* published on The Guild: <https://www.the-guild.eu/news-and-blog/blog/connecting-the-dots-iprs-knowledge-transfer-innova.html>

sustainability of research professions with the major aim of cultivating a highly skilled and versatile workforce capable of operating across academic, industrial, and societal interfaces (European Commission, 2022a).

- *ERA Action 14*, which underscores the importance of public engagement. This action promotes a shift towards more inclusive research ecosystems, encouraging stronger involvement of citizens and societal actors in the co-design and co-delivery of research and innovation. Such engagement is vital to ensure that R&I activities remain grounded in societal needs, ethical considerations, and democratic values (European Commission, 2022a).
- *The Horizon Europe Strategic Plan 2025–2027*, reinforcing these efforts by adopting a mission-oriented and impact-driven approach. It links the delivery of societal missions with ecosystem thinking and systemic innovation, by promoting the alignment of funding instruments, governance structures, and policy objectives. This strategic alignment leads to the conditions for more effective knowledge circulation and for scaling solutions that respond to complex, cross-sectoral challenges.

In this updated framing, knowledge valorisation is no longer understood as a linear or transactional process confined to the transfer of knowledge from academia to industry. Instead, it is viewed as a dynamic, iterative, and multi-actor process, embedded within broader systems of governance and interaction. It relies on diverse infrastructures, shared responsibilities across governance levels, and interdependent knowledge flows that span the entire research and innovation lifecycle.

3. MAPPING KNOWLEDGE VALORISATION IN EUROPEAN AND NATIONAL ECOSYSTEMS

The previous chapter outlined the EU's evolving policy framework for knowledge valorisation—highlighting the shift from fragmented, IP-centric approaches to integrated, systemic models guided by Open Science, mission-driven innovation, and the strategic management of intellectual assets. Building on that foundation, this chapter turns to the national level, where valorisation practices are translated, interpreted, and enacted through diverse governance structures, institutional capacities, and regional dynamics.

Rather than offering a benchmarking exercise or ranking national systems, this chapter adopts a systemic mapping approach to examine how each country mobilises knowledge for societal value. It focuses on the structural specificities, policy architectures, and actor configurations that shape national knowledge valorisation ecosystems. This is essential, because while the European Commission provides overarching principles, instruments, and coordination platforms, the implementation and effectiveness of valorisation strategies depend on localised pathways, enabling conditions, and institutional readiness. To this end, the chapter presents ecosystem snapshots of four countries—Italy, Spain, France, and Sweden—based on a combination of Mutual Learning Exercise (MLE) insights, national policy documents, and institutional case evidence shared by project partners. Each national section highlights:

- Strategic priorities and policy trajectories
- Institutional arrangements and governance mechanisms
- Key actors and stakeholder journeys
- Infrastructural enablers or barriers to systemic valorisation
- Relevant alignment or friction with EU frameworks.

The objective is not to provide exhaustive coverage but to identify actionable insights, common patterns, and systemic gaps that can inform ecosystem strengthening and transnational collaboration. These profiles offer a grounded basis for the development of readiness assessment tools and policy recommendations, as elaborated in later chapters.

Italy – National Knowledge Valorisation Landscape

Italy has transitioned from fragmented, IP-driven strategies toward a more systemic, EU-aligned approach to knowledge valorisation. Key reforms include the abolition of the professor's privilege (2023) and the revision of the Industrial Property Code (2024), shifting IP rights to institutions. While national frameworks like the National Recovery and Resilience Plan (NRRP) and the National Research Programme (PNR) integrate knowledge transfer across mission areas, a unified national strategy is still lacking, and governance remains distributed across ministries, agencies, and regions. Despite a strong research base, Italy faces coordination and absorptive capacity challenges. TTOs and KTOs exist in most universities, but their capabilities vary. ANVUR's VQR (National Agency for the Evaluation of Universities and Research Institutes developed Research Quality Assessment framework) increasingly includes third mission indicators and initiatives like ITA.CON (OECD, 2023) illustrating progress toward broader collaboration with societal actors. However, a culture of valorisation is still emerging, particularly outside elite institutions. The NRRP allocates €11B to research and innovation, supported by an additional €9.5B via the European Structural and Investment Funds (ESIF) and Smart Specialisation Strategies (S3). National programmes such as Smarter Italy, industrial PhDs, and competence centres promote public-private collaboration. Yet, structural gaps and limited long-term incentives hinder sustained institutional engagement. The ecosystem is marked by regional asymmetries, limited innovation procurement, and fragmented monitoring tools. Public administration often lacks capacity to act as a broker, and social innovation remains marginal. A cohesive culture of knowledge co-creation is still under development. Recent legal and policy reforms set the stage for a more integrated and mission-driven valorisation model. To advance, Italy must strengthen interoperable infrastructures, empower knowledge intermediaries, embed impact metrics, and align funding with long-term societal transitions. As summarised in Table 1, these developments reflect a gradual shift toward institutional maturity and closer alignment with EU-level strategic orientations for knowledge valorisation.

Table 1 - Italian R&I and Knowledge Valorisation Policy Timeline

Date	Policy/Action	Impact on Knowledge Valorisation
Late 1980s	Universities begin to develop internal policies for patents, spin-offs, consultancy	First steps toward institutional involvement in IP management
Pre-2000s	Fragmented national policies; limited coordination	Early foundations of academic entrepreneurship; no unified strategy
Early 2000s	Establishment of Technology Transfer Offices (TTOs)	Support for the commercialisation of research
2001–2005	"Professor's Privilege" introduced, then partially reversed	Researchers hold IP ownership initially; partial re-centralisation of institutions
2002	Founding of NETVAL (National Network for Research Valorisation)	Institutionalisation of national coordination for technology transfer
2010	Law 240/2010 introduces performance-based funding and external stakeholder participation	Boosts third mission visibility and strengthens university-society links
2011	ANVUR includes third mission in national evaluation (VQR)	Introduction of societal impact metrics
2017	Launch of Smarter Italy, competence centres, and industrial PhD initiatives	Territorial valorisation and applied research supported at a national scale
2021	National Research Programme (PNR) and National Recovery and Resilience Plan (NRRP) launched	Cross-cutting focus on impact, innovation ecosystems, and Open Science
2022	Italy joins CoARA; strengthens Open Science commitments	Alignment with EU-wide principles on research assessment and openness
2023 (May)	CNR (National Center for Research) adopts Open Science Roadmap	Institutional adoption of Open Science in the national research organisation
2023 (June)	Italian EOSC Tripartite Assembly (ITAEOSC-2023)	Multi-stakeholder governance for European Open Science Cloud
2023 (August)	Abolition of Professor's Privilege	Legal IP ownership moves to institutions; alignment with EU norms
2024	Revision of Industrial Property Code	Consolidates institutional valorisation responsibility
2024	Establishment of CoARA National Chapter (University of Bologna)	Reinforces Italian alignment with reform of research assessment and valorisation practices

Spain – National Knowledge Valorisation Landscape

Spain has progressively strengthened its legal and strategic commitment to knowledge valorisation, especially through the Science, Technology and Innovation Law (2022) and the State Strategy for Science, Technology and Innovation 2021–2027 (Gobierno de España, 2021). These frameworks embed valorisation within a broader agenda of sustainability, societal challenges, and public value creation—moving beyond a purely economic interpretation (European Commission, 2022a). Although the term "knowledge valorisation" is not always used explicitly, principles such as impact, co-creation, and transfer are prominent throughout national strategies. Coordination is led by the Ministry of Science and Innovation, with major roles played by the Centre for the Development of Industrial Technology (CDTI), Carlos III Health Institute (ISCIII), and the Spanish Foundation for Science and Technology (FECYT). However, inter-ministerial coordination remains fragmented, and unlike several other Member States, Spain has not adopted a dedicated national valorisation or innovation procurement strategy (OECD, 2023). Spain's ecosystem includes universities, public research centres, and intermediary bodies. Key institutions like CDTI, ISCIII, and FECYT promote collaborative R&I, open science, and citizen engagement. Most universities host TTOs or KTOs, though capabilities and maturity levels vary across regions. The 2022 reform of the Science Law has bolstered researcher mobility, cross-sectoral collaboration, and the recognition of societal impact in academic careers (Gobierno de España, 2022). Nonetheless, systematic institutional learning and knowledge-sharing mechanisms remain underdeveloped—especially outside frontrunner regions such as Catalonia, the Basque Country, and Madrid (European Commission, 2023). Spain leverages multiple funding streams to support knowledge valorisation. National initiatives such as the FID Line (Instrument for Promoting Innovation from Demand)) support pre-commercial and innovation procurement, while Research and Innovation Strategies for Smart Specialisation (RIS3) strategies shape regional investment. Spain's participation in Horizon Europe, national R&D programmes, and public-private partnerships (PERTEs) enrich its innovation portfolio (CDTI, 2023). Regional governments (e.g., Galicia, Andalusia) have piloted local innovation procurement programmes. Yet performance-based incentives remain limited, and many funding instruments still target individual projects rather than institutional transformation. Spain's valorisation landscape remains hindered by coordination bottlenecks, institutional fragmentation, and limited monitoring infrastructure. Despite the rhetorical commitment, the lack of a unified valorisation strategy, national spending targets, or harmonised definitions makes scaling and learning difficult (European Commission, 2022b). Regional disparities are significant, and while some areas lead in co-creation and innovation procurement, others face resource constraints and skills gaps. Within the university sector, valorisation remains a secondary priority, and cross-regional knowledge circulation occurs on an ad hoc basis. Spain has the infrastructure and policy frameworks to lead in participatory and demand-driven innovation. Moving forward, the focus must shift from policy proliferation to strategic coherence and systems-level governance. Strengthening inter-agency coordination, enhancing monitoring systems, and introducing structural incentives for institutional learning will be key. As summarised in Table 2, Spain's success will depend on building a learning ecosystem that integrates open science, citizen engagement, and innovation procurement into national and regional development strategies.

Table 2 - Spanish R&I and Knowledge Valorisation Policy Timeline

Date	Policy/Action	Impact on Knowledge Valorisation
2011	Creation of FECYT's open science and science culture programmes	Strengthened public engagement, citizen science, and open access infrastructures
2013	First generation of RIS3 (Smart Specialisation Strategies)	Framed regional innovation investment priorities and ecosystem development
2014	Establishment of PERTEs (Strategic Projects for Economic Recovery and Transformation)	Enabled large-scale public-private innovation projects in priority sectors
2017	Expansion of FID Line (Instrument for Innovative Demand)	Supported pre-commercial procurement and innovation-friendly public contracting
2020	National commitment to Open Science and FAIR data infrastructure	Boosted alignment with EU data sharing and interoperability standards
2021	Launch of State Strategy for Science, Technology and Innovation 2021–2027	Integrated impact, sustainability, and social challenges into national research policy
2022	Approval of the RIS3CAT 2030, Strategy for the smart specialization of Catalonia 2030	Framed regional innovation investment priorities and ecosystem development, addressing 7 shared agendas of Catalonia.
2022	Reform of the Science, Technology and Innovation Law	Strengthened research careers, mobility, and co-creation across sectors
2022	Spanish Universities and other organisations involved in the research assessment join CoARA and expand RRI principles in research evaluation	Aligns national frameworks with EU priorities on societal impact and responsible research
2023	Regional innovation procurement pilots in Galicia and Andalusia	Tested local valorisation tools and innovation-oriented purchasing mechanisms
2024	Creation of the National Office for Scientific Advice (ONAC)	One of the main goals of ONAC is to ease the access of the General State Administration to scientific knowledge for the design of public policies.
2024	Development of monitoring frameworks for national science and innovation strategies	First steps toward performance-based tracking of knowledge transfer and societal impact

France – National Knowledge Valorisation Landscape

France has a long tradition of embedding innovation and knowledge valorisation in its national policy, with the state playing a central steering role (Cour des Comptes, 2022). Strategic frameworks such as the Loi de Programmation de la Recherche (2020), France 2030, and the National Strategy for Research and Innovation (SNRI) provide strong top-down direction for research-industry collaboration, technology transfer, and innovation diffusion (MESRI, 2020; France Stratégie, 2021). While tools such as the Public Procurement Code allow for innovation partnerships, the legal framing of innovation procurement diverges from EU definitions (OECD, 2023). Knowledge valorisation is often equated with IP commercialisation and start-up creation, with limited institutional emphasis on open knowledge practices or societal innovation (European Commission, 2022a). France hosts a dense landscape of actors, including public research organisations (e.g. CNRS, INRAE, Inserm), universities, national agencies, and state-backed technology transfer bodies such as the SATTs. These are coordinated by central ministries and regional authorities, supported by specialised bodies like the DAE (public procurement) and AID (defence innovation) (Cour des Comptes, 2022). The SATT system plays a pivotal role in IP valorisation, though its integration into wider KEC and societal impact frameworks remains partial. Regional disparities persist: while regions like Nouvelle-Aquitaine are proactive in Smart Specialisation and open science, others remain peripheral to national innovation policy. France offers a diverse array of R&I funding tools—national (e.g. France 2030, Bpifrance), regional, and thematic. Tax credits such as the Crédit d’Impôt Recherche (CIR) (Bpifrance, 2023) incentivise private R&D investment, and initiatives like French Tech 2030 aim to double innovative SME presence in public procurement by 2027 (Bpifrance, 2023). However, these mechanisms prioritise the supply side of innovation (start-ups, patents, IP), with fewer institutional incentives for public sector co-creation, knowledge commons, or social value generation (OECD, 2023). Recognition and capacity-building for third mission engagement in HEIs remain limited. Despite strong central governance, France faces coordination challenges across ministries and levels of government. The absence of a dedicated national strategy for knowledge valorisation results in fragmented implementation. Local institutions often lack autonomy or incentive frameworks to drive societal engagement. Innovation procurement remains uneven and concentrated at the national level. Training and awareness programmes (e.g. via CNFPT, IGPDE) are underutilised and do not consistently reach public research and academic institutions (European Commission, 2022b). France has the institutional infrastructure and policy ambition to lead in EU knowledge valorisation. Unlocking this potential will require investing in horizontal governance, empowering regional ecosystems, and embedding valorisation into the mission of public institutions. As summarised in Table 3, France must now pivot from innovation as procurement to innovation as co-creation—anchoring its strong central institutions in a more inclusive and interoperable valorisation ecosystem.

Table 3 - French R&I and Knowledge Valorisation Policy Timeline

Date	Policy/Action	Impact on Knowledge Valorisation
1967	ANVAR (Agence nationale de valorisation de la recherche)	Created to support the valorisation of public research. In 2005, it merged with other bodies to form Oséo.
1994-1996	Pierre Potier’s tenure as Director General of Research and Technology	Proposed key decrees (1995–1996) granting bonuses to researchers involved in the commercialisation of their inventions.
1998	Henri Guillaume report	Highlighted the weak ties between public research and socio-economic actors in France.
1999	Loi Allègre (Allègre Law)	Launched under Minister Claude Allègre, it enabled the creation of public research incubators and the i-LAB competition for innovative startups, supported by ANVAR.
2006	Creation of Pôles de Compétitivité (Competitiveness Clusters)	Launched cross-sector innovation ecosystems across regions
2009	Establishment of Investissements d’Avenir (PIA) programme	Long-term strategic investment in R&D and innovation infrastructure
2010–2014	Formation of SATTs (Sociétés d’Accélération du Transfert de Technologies)	Structuring of regional IP and spin-off commercialisation system
2013	Launch of the National Strategy for Research and Innovation (SNRI)	Aligned research and innovation policy with EU grand challenges
2015	Introduction of the Public Procurement Code (including innovation partnership clauses)	Opened legal space for experimental procurement and innovation clauses
2019	DAE and AID gain visibility as specialised procurement and innovation agencies	Reinforced thematic valorisation strategies in defence and public procurement
2020	Loi de Programmation de la Recherche (LPR)	Institutionalised long-term planning for research and valorisation
2021	Launch of France 2030 investment plan	Mobilised €54B over 5 years for green, digital, and industrial innovation missions
2021	Expansion of Crédit d’Impôt Recherche (CIR)	Reinforced tax incentives for private sector research
2022	National mobilisation for Open Science and research data management	Growing emphasis on data interoperability and scientific openness
2023	Launch of French Tech 2030 and updated I Choose French Tech initiative	Target to double innovative SMEs in public procurement by 2027
2023	French Universities and other organisations involved in the research assessment join CoARA	Aligns national frameworks with EU priorities on societal impact and responsible research
2024	National roadmap for innovation procurement under review	Ongoing effort to align national legal definitions with EU valorisation principles

Sweden – National Knowledge Valorisation Landscape

Sweden is internationally recognised for its innovation-oriented culture, high public R&D investment, and commitment to evidence-based policymaking (OECD, 2023). National frameworks such as the Swedish Innovation Strategy (Swedish Government, 2012), the Digitalisation Strategy, and the Smart Industry Strategy define valorisation as an enabler of societal transformation (Swedish Government, 2012; Vinnova, 2019). Governance is distributed across public agencies, municipalities, and innovation programmes, with an emphasis on enabling conditions—such as regulatory flexibility, experimentation, and capacity-building—over centralised control. While this decentralised model fosters institutional autonomy and innovation agility, it results in limited strategic coherence around knowledge valorisation and underdeveloped formal policies for innovation procurement and systemic knowledge sharing (OECD, 2023). Sweden’s ecosystem includes universities, public authorities (notably VINNOVA, the National Agency for Public Procurement), regional councils, and multistakeholder platforms (OECD, 2023). VINNOVA – the Swedish Innovation Agency – plays a leading role in promoting challenge-driven innovation, testbeds, and open science infrastructures. The launch of Afori—a national platform for innovation procurement practitioners—marks a step toward cross-sectoral coordination (Vinnova, 2023). Yet, despite widespread engagement in innovation by universities and municipalities, valorisation maturity varies significantly, and the absence of national benchmarking or coordinated frameworks limits systemic progression. Sweden uses both national and EU funds to support collaborative innovation. VINNOVA issues competitive calls for public-private partnerships, mission-driven projects, and early-stage R&D. The Strategic Innovation Programmes (SIPs)—co-funded by government and industry—target key domains like mobility, energy, and health (Vinnova, 2021). However, funding for knowledge valorisation is fragmented, often indirect, and lacks dedicated budget lines. Sweden’s decentralised governance supports experimentation but complicates scaling. Autonomy at the municipal level enables local adaptation but leads to heterogeneous practices and limited policy diffusion. Open science enjoys normative support but lacks operational consistency. The absence of shared metrics, capacity-building pathways, and systemic incentives hampers broader uptake. Institutional inertia, fragmented coordination, and the absence of personal incentives for co-creation remain key barriers to transformation (European Commission, 2022). Sweden is well-positioned to lead in distributed, participatory models of knowledge valorisation. A culture of public trust, digital openness, and cross-sector experimentation provides a strong foundation. The challenge now is to connect isolated excellence through shared frameworks—embedding valorisation across governance levels and aligning it with societal missions. As summarised in Table x, Sweden’s next step is to evolve from decentralised experimentation to structured learning and collective impact—recognising knowledge not only as a driver of innovation but as a public good.

Table 4 - Swedish R&I and Knowledge Valorisation Policy Timeline

Date	Policy/Action	Impact on Knowledge Valorisation
2001	Establishment of VINNOVA (Swedish Innovation Agency)	Central innovation agency promoting collaborative R&I and public-private partnerships
2012	Launch of the Swedish Innovation Strategy	National strategy linking innovation to societal challenges and systemic transformation
2015	Introduction of the Digitalisation Strategy	Framework for data-driven innovation, public sector experimentation, and digital capacity-building
2016	Establishment of Strategic Innovation Programmes (SIPs)	Co-funded by government and industry; targets mission areas like health, climate, and mobility
2017	Publication of Smart Industry Strategy	Focus on industrial transformation through digitalisation, AI, and innovation ecosystems
2018	Introduction of VINNOVA's Challenge-driven Innovation funding schemes	Promotes cross-sector collaboration and societal impact
2021	Expansion of open science and FAIR data infrastructures	Supports research data sharing, interoperability, and citizen science
2022	Launch of Afori (Innovation Procurement Community of Practice)	Strengthens practitioner coordination and public sector innovation capabilities
2023	National discussions on innovation procurement roadmap and reward systems	Emerging focus on institutional incentives and systemic valorisation beyond tech transfer
2023	Swedish Universities and other organisations involved in the research assessment join CoARA	Aligns national frameworks with EU priorities on societal impact and responsible research

Despite only Spain and Sweden being among the countries who participated in the Mutual Learning Exercises for Knowledge Valorisation, a generalised analysis of MLE reports¹³—conducted through desk-based comparative research—offered us a structural overview of how national-level KV ecosystems function across different contexts. Each country case documented in the MLEs serves as a situated configuration¹⁴, enabling a comparative understanding of system-specific challenges and enablers. From these reports, it is possible to abstract a recurring set of actor categories—such as higher education institutions (HEIs), public research organisations, ministries, regional development agencies, intermediaries, industry stakeholders, and societal actors. These actors are embedded in complex value flows, including the circulation of knowledge, funding, personnel, regulatory mechanisms, physical and digital infrastructure, social legitimacy, and intellectual property rights (European Commission, 2024). By analysing these interactions through an information architecture and systemic design lens, common actor networks and recurring flow patterns begin to emerge. At the same time, structural asymmetries become visible—for example, the persistent lack of strong engagement with civil society actors, or limited alignment between national and EU-level funding streams and priorities. This type of systemic mapping reveals three key layers of insight:

- The infrastructural conditions that shape or constrain knowledge flows (e.g., legal frameworks, funding mechanisms, regional innovation strategies).
- The institutional readiness and behavioural capacities of actors to engage in co-creation, knowledge sharing, and transfer.
- The misalignments between policy intent and implementation mechanisms, which can generate inefficiencies, gaps, or conflicting incentives across systems (European Commission, 2024).

These insights have guided the development of the tool and framework introduced in the following chapter, designed to be actor-specific while sensitive to broader system interdependencies. Together, they form the analytical basis for identifying different levels of knowledge valorisation readiness—understood here as the degree to which actors and ecosystems are equipped to participate meaningfully in collaborative, value-generating processes. Readiness is thus a function of infrastructural support, institutional culture and behaviour, and the capacity to maintain collaborative flows over time.

¹³ Mutual Learning Exercises (MLEs) are structured peer-learning processes initiated by the European Commission under the Horizon Europe Policy Support Facility. Since their launch in 2016, MLEs have enabled Member States and associated countries to exchange experiences, compare policy approaches, and co-develop recommendations on strategic topics related to research and innovation policy. Each MLE results in a final report that synthesises key findings, national case studies, and policy lessons.

¹⁴ We use the term situated to describe knowledge valorisation processes that are embedded in specific socio-technical and institutional contexts. For more, see Haraway(1988)¹

4. KNOWLEDGE VALORISATION READINESS SELF - ASSESSMENT TOOL

As discussed in the previous chapters, we adopt an ecosystem approach to knowledge valorisation, which refers to a model that fosters collaboration among stakeholders, regional authorities, and central governments.

Building on such an approach to knowledge valorisation, we developed a self-assessment tool designed to help Higher Education Institutions (HEIs) identify their current approach and development in terms of knowledge valorisation. We rooted the self-assessment framework in a multidimensional conception of knowledge valorisation, which moves beyond traditional technology transfer. It embraces the active contribution of all societal actors—including citizens and public authorities—in co-creating value from knowledge, in line with the broader vision articulated in the R&I Channel EU Policy Brief¹⁵. The tool outputs a “readiness level” that assesses the maturity of the organisation in relation to the knowledge valorisation activities. According to the Engagement Readiness Monitor¹⁶, the concept of “readiness” in higher education institutions (HEIs) encompasses two interdependent dimensions: preparedness and willingness. Preparedness refers to the institutional capacity to engage in cooperation, including the presence of appropriate resources, competencies, structures, and support mechanisms that enable collaboration if desired. In contrast, willingness relates to the institution’s inclination or favourable disposition toward external engagement, expressed through organisational behaviours, values, leadership attitudes, and strategic orientation. Both elements are essential to determine whether an HEI is effectively positioned to establish and sustain impactful partnerships with external stakeholders.

Our toolkit enables self-assessment across the dynamics of knowledge flows within the six R&I Valorisation Channels as outlined in the relevant European Commission Policy Brief.

Starting from this framework, we have integrated the concept of “knowledge share” (see Box 2). According to Grimaldi and Conti (2024), the notion of Knowledge Share goes beyond the traditional concepts of knowledge “transfer” and “exchange.” It captures a broader and more dynamic process that unfolds through both intentional and unintentional flows, enabling distribution, connection, and co-production of knowledge. Unlike linear and monodirectional models, knowledge share operates as a circular, multi-sourced, and multi-directional system. Knowledge can emerge from various sources, evolve iteratively through diverse actors’ interactions, and often generate unanticipated outcomes. Importantly, it also encompasses the socialization of tacit knowledge, making it available, accessible, and meaningful through interaction and trust-building. This shift allows us to reframe how knowledge flows are understood and assessed—emphasizing not just the endpoints of transfer, but the complex, emergent, and embedded nature of knowledge circulation across environmental, organisational, and individual levels.

In the following paragraphs, we introduce each element composing the knowledge valorisation self-assessment tool, namely: the actors participating in the process, the channels in which they interact, and the enabling factors, which are the elements influencing the knowledge flows.

✱ Actors

The actors participating in the process of knowledge valorisation reflect a Quadruple Helix configuration, engaging distinct but interconnected systems of knowledge production, application, and transformation. These include:

1. **Academia**, encompassing universities, research-performing organisations, and technology transfer offices, as primary producers and stewards of knowledge and innovation.
2. **Industry**, including SMEs, large corporations, and start-ups, acting as both users and co-developers of knowledge, contributing to market uptake, scaling, and sectoral transformation.
3. **Private investors**, such as banks, venture capital funds, and business angels, who mobilise financial capital and de-risk early-stage innovation, playing a critical role in the valorisation pipeline.
4. **Public authorities**, including local, regional, national, and EU-level institutions and funding agencies, which shape enabling policy environments and steer strategic investment through instruments like Smart Specialisation Strategies and European Structural and Investment Funds (ESIF).
5. **Civil society**, comprising NGOs, user groups, and communities, which brings in diverse epistemologies, ensures societal relevance and is key to responsible innovation and inclusive impact generation.

This asset of actors not only channels knowledge across institutional boundaries but also supports the distributed, circular, and multi-actor nature of knowledge share, making valorisation a deeply relational and systemic process.

¹⁵ European Commission: Directorate-General for Research and Innovation, Research & innovation valorisation channels and tools – Boosting the transformation of knowledge into new sustainable solutions, Publications Office, 2020, <https://data.europa.eu/doi/10.2777/480584>

¹⁶ The Engagement Readiness Monitor Synthesis Report (2021), accessible at https://engagementready.eu/wp-content/uploads/2022/12/Engagement-Readiness-Monitor_IOI_Synthesis-Report_FINAL-24112021_Corrected.pdf

✳ Channels¹⁷

Our tool enables self-assessment across the dynamics of knowledge flows within the six Research & Innovation (R&I) Valorisation Channels as outlined in the relative European Commission Policy Brief¹⁸. These channels are visible in Box 1.

Box 1 – Knowledge Valorisation Channels description

▫ Channel 1 | Academia-Industry Joint Research & Mobility

Joint research and inter-sectoral mobility play a strategic role in the valorisation of R&I fostering a mutual exchange between industry and academia. On the one hand, they facilitate the flow of academic knowledge and results into industries. On the other hand, they offer researchers from academia the opportunity to enhance their skills and gain better knowledge of industrial needs.

▫ Channel 2 | Promoting research-driven spin-offs and start-ups (access to financing)

The promotion of spin-offs and start-ups is key for the valorisation of R&I results as they offer students and academics an entrepreneurial route to commercialising the knowledge developed. In this context, a crucial role is played by structured access to finance for these early-stage companies as it creates the framework conditions needed to drive the transformation of research results into products and services.

▫ Channel 3 | Intermediaries and knowledge transfer professionals

Intermediary organisations – such as knowledge transfer offices (KTOs), technology transfer offices (TTOs), business incubators and science parks – create a channel for knowledge valorisation by helping researchers and innovators commercialise their solutions, products and services. They improve the conditions for knowledge transfer by serving as the first contact point for researchers and industry looking for new opportunities. They also promote additional instruments and services to boost the innovation potential of research through networking, mentoring activities, coaching and exchange of best practices

▫ Channel 4 | Engagement of citizens, public bodies and societal actors

Citizen engagement is a major way to ensure that new knowledge will not only lead to innovative solutions but will result in those that matter to citizens. Such solutions have a higher societal acceptance and therefore are also more effective. When citizens are in the driving seat and in a co-creation process, knowledge has a high potential for societal uptake.

▫ Channel 5 | Intellectual property management & “standardisation”

The valorisation of research results is facilitated by a fit-for-purpose intellectual property (IP) framework which assures the creators of research outputs a competitive advantage in the business. Standards also have a strong value for knowledge valorisation, facilitating the access of new products to the market.

▫ Channel 6 | Knowledge circulation & informing policy

Knowledge dissemination and the policy uptake of research results offer another channel for knowledge valorisation. Technology and science are becoming increasingly open, global, interdisciplinary and digital. Ensuring that advances in science and technology are as open as possible is key in our knowledge-driven world where data are increasingly valuable and access to data is considered a competitive advantage. Research results, such as studies, data, models, experiments, and theoretical analyses, are of the greatest benefit to data-informed policymaking. They can help decision-makers better understand the nature of the challenges they are facing and the possible implications of the decisions they may make.

✳ Enabling factors

The knowledge valorisation practices have been therefore analysed through three enabling factors: environmental, organisational, and individual factors. These categories, as described by Grimaldi and Conti (2023), provide a comprehensive framework for understanding the multilevel conditions that shape “knowledge sharing” processes within and beyond higher education institutions:

- **Environmental** factors refer to the broader contextual elements—such as regulatory frameworks, political priorities, and socio-economic trends—that influence how knowledge is mobilised.
- **Organisational** factors encompass the internal structures, policies, resources, and routines that support or hinder knowledge valorisation.
- **Individual** factors concern the behaviours, attitudes, and capabilities of the people involved—researchers, students, staff, and external partners—who ultimately enact and sustain knowledge sharing. By adopting this framework, we aim to capture the systemic, institutional, and human dimensions that underpin knowledge valorisation readiness.

¹⁷ The following content is presented as displayed in: European Commission: Directorate-General for Research and Innovation, Research & innovation valorisation channels and tools – Boosting the transformation of knowledge into new sustainable solutions, Publications Office, 2020, <https://data.europa.eu/doi/10.2777/480584>

¹⁸ European Commission: Directorate-General for Research and Innovation, Research & innovation valorisation channels and tools – Boosting the transformation of knowledge into new sustainable solutions, Publications Office, 2020, <https://data.europa.eu/doi/10.2777/480584>

Box 2 – Description of Knowledge Share concept¹⁹

Specifically, sharing knowledge means distributing both explicit and tacit knowledge, socializing it and making it available and accessible. Sharing knowledge means connecting and bringing together different content, information, pieces of knowledge and people and/or organizations.

Sharing knowledge means co-producing, leveraging others' knowledge contents and bringing them together to build new ones. KS enables [...] us to bring together data/information/knowledge from different sources (connecting) to favour joint generation (co-producing) of new knowledge that is socialized and diffused (distributing).

In the Knowledge Share domain, knowledge is not created in a specific moment (time) or context (space); it is created through a process in which an initial spark of knowledge is socialized within a broader context, where it is enriched by different perspectives, as various contributors come into play to voluntarily or involuntarily co-produce.

This multidimensional mapping has the aim to capture knowledge valorisation dynamics within the organisation, i.e., HEI. By analysing channels through environmental, organisational, and individual lenses, we highlight the need for universities—key producers of knowledge in society—to recognise that knowledge is increasingly created in dynamic, distributed, and often informal ways.

✳ Readiness Levels

The output of the self-assessment provides a framework for situating the institution within one of three progressive levels of readiness in relation to knowledge valorisation capabilities:

A) LOW

At this foundational stage, Higher Education Institutions (HEIs) and Public Research Organisations (PROs) demonstrate a basic recognition of the internal and external drivers that shape the movement, use, and transformation of knowledge. This includes emerging awareness of stakeholder needs, policy instruments, and knowledge-sharing mechanisms, although concrete practices may remain limited or informal.

B) MEDIUM

Institutions at this level begin to operationalise their awareness through the establishment and structuring of specific channels for knowledge valorisation. This may involve the formalisation of roles (e.g. knowledge transfer offices), the creation of internal coordination mechanisms, or the implementation of targeted activities such as spin-off support, citizen engagement, or research-policy interfaces. Interactions are more deliberate, yet still largely domain-specific or siloed.

C) HIGH

At the highest level of readiness, institutions exhibit an integrated, systemic approach to knowledge valorisation. Channels are not only developed but also strategically aligned across organisational units and actor ecosystems. Practices are embedded within institutional governance, supported by reflexive evaluation mechanisms, and framed by long-term capacity-building strategies. At this stage, institutions tend to act as orchestrators of multi-actor knowledge ecosystems, embracing co-creation, engagement, open science principles, and inclusive innovation.

The full Knowledge Valorisation Readiness Self-Assessment Tool is available in Annex I, which includes a detailed set of diagnostic questions designed to assess the HEI's maturity across organisational, environmental, and personal dimensions. The tool supports both internal reflection and cross-institutional comparison within the Quadruple Helix framework, offering an entry point for strategic capacity building.

To assess and test the tool, The University of Bologna has been selected as the “case zero” for the initial mapping of knowledge valorisation practices, providing a concrete context to test and refine the self-assessment methodology. As the lead partner for this Work Package, its ecosystem offers a valuable reference point for exploring how knowledge flows across the six knowledge valorisation channels and how these are influenced by the three enabling factors. Our preliminary analysis establishes a baseline for future comparative exercises across the consortium, drawing on both qualitative and quantitative insights.

¹⁹ As it appears on Conti, G., & Grimaldi, R. (2024). Knowledge Share: The (R)evolution of Technology Transfer. Springer Cham. <https://doi.org/10.1007/978-3-031-51384-8>

5. POLICY GUIDELINES

Despite increasing emphasis on the interactions between open and closed science approaches for knowledge valorisation, there remains limited empirical insight into how stakeholders function within real-world valorisation systems. Studies of innovation ecosystems have highlighted how existing models often underplay the micro-dynamics of coordination, value creation, and actor interdependence within the Quadruple Helix framework (Yun and Liu, 2019). These imbalances may privilege institutional incumbents and underrepresent emergent or peripheral actors. To address this, future policy design must move beyond compliance metrics and static typologies toward tools that reflect the fluid, contested, and adaptive nature of knowledge systems. This includes adopting socio-technical diagnostics such as crisis resilience analysis, collective intelligence frameworks, participatory simulation, and mixed-method system mapping (Hasche et al., 2020). Recent methodological contributions suggest that policy instruments should be designed not only to incentivise behaviours, but to model how knowledge circulates, mutates, and adapts in dynamic environments. This involves understanding how actors co-evolve with systems, interpret institutional signals, and reshape their roles through feedback-rich, uncertain, or crisis-prone contexts (Nespeca et al., 2023). From a policy design standpoint, this implies:

- Embedding reflexive monitoring tools that blend qualitative and quantitative insights.
- Developing modular, iterative evaluation frameworks.
- Supporting experimental governance infrastructures — such as policy labs, digital twins, or foresight-driven living labs (Kalenov & Shavina, 2018).

These approaches make it possible to operationalise knowledge valorisation not as a linear process, but as a conditionally co-produced outcome — one that reflects collective learning and contextual adaptation rather than fixed institutional templates (Nespeca et al., 2024). In the next sub-chapter, we outline ten actionable policy guidelines aimed at operationalising knowledge valorisation across diverse institutional and ecosystem contexts. These guidelines, presented in Table 5, are grounded in empirical insights from the project’s diagnostic tools, pilot experiences, and stakeholder consultations. They reflect the need to move beyond declarative commitments toward structural alignment — particularly by addressing incentive systems, actor configurations, and institutional readiness. While they can be adapted to different maturity levels, their collective aim is to support more inclusive, reflexive, and sustainable knowledge ecosystems in line with the EU’s strategic goals.

5.1 GUIDELINES

The following guidelines are derived from the KALEIDOS project’s cross-cutting analysis of institutional dynamics, ecosystem configurations, and the operationalisation of the EU Guiding Principles for Knowledge Valorisation. Building on the multidimensional prospect gained, they aim to provide strategic orientation for policymakers, university leaders, intermediaries, and other actors involved in knowledge valorisation across Europe.

Rather than opting for one-size-fits-all solutions, the guidelines reflect a systemic view of valorisation as a multi-actor, mission-oriented process. They are grounded in the need to diagnose institutional readiness, rebalance actor roles, and sustain long-term ecosystem collaboration. Aligned with the EU Knowledge Valorisation Policy Toolbox, this section emphasises the importance of adaptable tools, co-creation mechanisms, and enabling conditions. To enhance clarity and practical uptake, the policy guidelines have been grouped according to the environmental, organisational, and individual factors that shape knowledge-sharing ecosystems. This structured approach helps identify actionable levers for supporting ethical, inclusive, and context-responsive value creation from knowledge.

Building on existing European Commission frameworks and instruments for knowledge valorisation, the following table presents ten actionable policy guidelines organised across key thematic areas. Each guideline articulates a clear vision aligned with EU strategic priorities, identifies persistent bottlenecks—whether institutional, systemic, or structural—and proposes targeted recommendations to address them. Rather than prescribing generic solutions, the guidelines aim to strengthen institutional capacities, recalibrate incentive systems, and embed reflexive, mission-oriented governance. The table provides a practical framework to support more coherent and context-sensitive implementation of EU knowledge valorisation ambitions.

Table 5 – Policy Guidelines for Knowledge Valorisation

POLICY AREA	VISION	BOTTLENECKS	RECOMMENDATION	Level of analysis
Institutional Readiness and Self-Assessment	Every university has the capacity to assess, plan, and act on its valorisation mission.	Institutional readiness is not static. A maturity-based portfolio approach is needed to align capacities with multiple policy objectives. Valorisation strategies must reflect internal diagnostics — not only to “select” promising initiatives but to compose balanced and context-responsive portfolios of engagement. This helps shift from ad-hoc efforts to strategic, transparent, and long-term institutional strategy planning.	1. Implement Diagnostic Tools: Deploy maturity models and readiness frameworks as standard policy instruments. 2. Design Portfolio-Based Action Plans: Move beyond single ‘best-bet’ projects. Build tailored strategies that combine complementary valorisation modes aligned with mission goals, risk profiles, and capacity levels.	<i>Organisational, Environmental</i>
Incentives for Researcher Engagement	A European research system where valorisation is embedded in academic careers and reward structures.	Valorisation remains aspirational without structural incentives. Academic evaluations still favour publication metrics over co-creation, societal impact, or engagement.	3. Align Evaluation Systems: Integrate engagement and impact in tenure, grant, and promotion criteria. 4. Reform Career Pathways: Recognise diverse academic contributions including public engagement, interdisciplinary teaching, and entrepreneurial activities.	<i>Individual</i>
Systemic and Socio-Technical Tools	Polymaking and valorisation are guided by systems thinking and adaptive foresight.	Static models are insufficient. Complex ecosystems require dynamic, qualitative tools to map uncertainty, manage change, and guide adaptive governance.	5. Fund Reflexive Infrastructure: Invest in scenario labs, systems mapping and participatory simulations to support strategic adaptability. Fund longitudinal strategic reviews and differentiated performance indicators to recognise diverse pathways to impact.	<i>Environmental, Organisational</i>
Actor Configuration and Power Dynamics	Balanced ecosystems where emergent actors can shape agendas and access resources.	Valorisation ecosystems are often distorted by legacy hierarchies. Universities, intermediaries, and civic actors operate with mismatched power and incentives.	6. Rebalance Actor Roles: Support shared governance, fund civic and intermediary participation, and foster inclusive agenda-setting.	<i>Organisational</i>
Intermediaries and Knowledge Brokers	Intermediaries are recognised as strategic actors in shaping valorisation processes.	Innovation brokers, research managers, and KTOs do more than support — they interpret, translate, and connect knowledge across boundaries. Yet their roles remain underfunded and undervalued.	7. Strengthen Intermediaries: Create dedicated funding lines, long-term positions, and capacity-building programmes for intermediaries and boundary-spanners.	<i>Individual, Environmental</i>
Strategic Orientation and Institutional Profiling	Universities contribute to knowledge valorisation through differentiated, mission-aligned strategies that reflect their identity, capabilities, and territorial roles.	Current policies often assume homogeneity in university missions and valorisation strategies. This leads to mismatches between institutional capabilities and policy expectations, particularly when funding, evaluation, or support instruments fail to account for strategic diversity across the higher education system.	8. Support Strategic Profiling: Encourage universities to articulate their valorisation strategies based on their identity (e.g. generalist/specialist, high/low prestige, regional/global scope). Tailor policy instruments to reward alignment between mission, capability, and societal contribution — not just commercial outputs.	<i>Environmental, Individual</i>
Mission Coherence and Local Anchoring	Universities and ecosystems align knowledge valorisation with place-based needs, fostering long-term public value through territorial anchoring.	Fragmented coordination between national and local actors weakens the effectiveness of mission-oriented innovation. Universities may lack incentives or institutional mechanisms to embed valorisation in local development priorities.	9. Align Missions with Local Capacities: Design policy frameworks that incentivise universities to act as anchor institutions in regional ecosystems. Encourage co-development with local public actors, SMEs, and civic organisations. Provide resources for place-based experimentation and adaptive governance tailored to territorial priorities.	<i>Organisational, Environmental</i>
Learning and Peer Reflection	Knowledge ecosystems evolve through situated learning, iteration, and openness to failure.	Reflexivity and shared learning are undervalued. Without capturing failures and context-specific insights, peer learning remains superficial.	10. Institutionalise Peer Learning: Create modular tools and open-access repositories to document, compare, and learn from valorisation experiences.	<i>Individual</i>

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✳ Sitography

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- Piano Nazionale di Ripresa e Resilienza (PNRR) <https://www.governo.it>
- French Tech 2030 & Innovation Procurement – Bpifrance <https://bpifrance.fr>
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- EU Research and Innovation Policy Portal (Knowledge Valorisation Section) <https://research-and-innovation.ec.europa.eu>
- SSH Centre – Citizen Engagement in EU Initiatives <https://sshcentre.eu>
- Vinnova – Swedish Platform for Innovation and Research <https://www.vinnova.se>

ANNEX I - Knowledge Valorisation Readiness Levels

To operationalise knowledge valorisation readiness across Higher Education Institutions (HEIs), we developed a self-assessment tool that maps key practices at the intersection of actors, valorisation channels, and systemic factors—namely, the environmental, organisational, and individual dimensions²⁰. The self-assessment tool is designed to support structured reflection, internal dialogue institutional diagnosis, strategic alignment, and cross-actor dialogue-

The tool is built around six EU-defined R&I knowledge valorisation channels and includes tailored diagnostic questions for each. While each question is linked to a specific channel, the three systemic dimensions of knowledge sharing—environmental (ecosystems, partnerships, policies), organisational (structures, routines, governance), and individual (skills, incentives, behaviours)—are implicitly embedded in the logic of the tool.

✳ Understanding the Readiness Levels

The self-assessment tool is built around two types of diagnostic questions; each aligned with a specific function. First, closed-format mapping questions (Yes/No) identify the presence or absence of key structures, roles, or mechanisms within the institution. These aspects help map actor configurations and governance components. Secondly, qualitative questions use a 1–7 Likert scale to assess the systemic quality, coherence, and overall maturity of knowledge valorisation practices. These questions allow institutions to reflect on how well knowledge flows are embedded, supported, and evolving. These two formats capture both the structural and dynamic aspects of KV readiness, enabling institutions to move from descriptive diagnosis to strategic development.

All responses in the self-assessment tool are scored on a 1–7 scale. Scores are averaged per dimension (environmental, organisational, individual) within each valorisation channel. These partial scores are then used to determine a channel-level score, and eventually an overall institutional score. The readiness levels—Low (1.0–3.4), Medium (3.5–5.4), and High (5.5–7.0)—help institutions interpret the maturity of their practices at each level of analysis.

These levels serve as interpretive anchors to help institutions be aware of where they stand—not only within individual channels, but also across systemic dimensions. They indicate whether practices are still emerging, already coordinated, or fully institutionalised. Because there is no one-size-fits-all model for knowledge valorisation, the tool is deliberately flexible. Used over time and across organisational units, it can help track progress, foster dialogue between actors, and support data-informed decision-making. Rather than enforcing standardised benchmarking, the tool promotes a shared diagnostic culture that supports more intentional, inclusive, and strategic knowledge valorisation.

✳ Pre-Assessment Mapping and Reflection

Before engaging directly with the readiness self-assessment tool, institutions are strongly encouraged to carry out a preliminary mapping of their knowledge valorisation ecosystem. This initial reflection ensures that respondents have a clear understanding of the actors, structures, and dynamics at play within each KV valorisation channel. Approaching the tool without this overview may limit its usefulness, as many questions require familiarity with who is involved in specific processes, where responsibilities lie, and what institutional mechanisms are in place. To support structured reflection, institutions are encouraged to explore how their knowledge valorisation ecosystem is internally configured by mapping each of the six channels in their own context. This exercise helps visualise how different actors—academia, industry, government, and civil society—operate across channels, and how environmental, organisational, and individual factors shape their contributions. To guide this process, we provide a set of preparatory questions (Box 3), which can be used in internal workshops or consultations to clarify roles, identify gaps, and build a shared institutional understanding before completing the assessment.

Box 3 – Pre-Assessment guiding questions

- How is this channel defined in our institution?
- Do we share a common understanding of what this channel includes or excludes?
- Who operates in this channel in our context (academia, industry, government, civil society)?
- Are there missing or under-engaged actor groups?
- Who leads coordination? Who initiates the activity? Who benefits?
- What structures or units support this channel (offices, labs, programs, working groups)?
- Are these structures stable, cross-functional, and well-connected to other channels?
- How does regional, national, or EU policy influence this channel in our institution?
- Are there external incentives or funding schemes we rely on?
- Who are the key individuals active in this channel (profiles, roles, training)?
- Are their contributions recognised in evaluation or career progression?
- Do we consider this channel “institutionalised”, “in development”, or “ad hoc”?
- What evidence supports our answer (strategic plan, metrics, internal reviews)?
- Which dimensions (environmental, organisational, individual) are most or least developed?

²⁰ Identified by Grimaldi and Conti (2023).

✴ Using the Tool: Recommended Contributors and Timeframe

The time required to complete the tool may vary depending on the stage and purpose of use. For institutions undertaking the initial mapping of their knowledge valorisation ecosystem (e.g. identifying actors, structures, and channel configurations), the full process may require up to 2.5–3 hours, particularly if it is done collaboratively across units. In contrast, if the mapping has already been completed and the focus is only on completing the KV self-assessment tool, institutions can expect to allocate approximately 30 to 60 minutes, depending on the level of internal coordination and familiarity with the channels. To ensure accurate and meaningful responses, the readiness tool should ideally be completed by a cross-functional group. This might include staff from the Knowledge Transfer Office, Open Science coordinators or Data Stewards, academic leaders, and relevant administrative or policy personnel. In contexts where citizen engagement or societal impact is assessed, it may also be useful to involve staff responsible for public engagement or civic science initiatives. The composition of the group can be adapted depending on whether the focus is a full ecosystem mapping or a more targeted self-assessment.

Table 1a presents the full set of diagnostic questions characterising the self-assessment tool. Organised by valorisation channels and relative enabling factors (environmental, organisational, individual), the questions are designed to help institutions evaluate both the structural presence of key actors and mechanisms and the systemic maturity of their knowledge valorisation practices. Each question contributes to generating a readiness score that supports reflection, benchmarking, and strategic planning. Finally, an applied version of Table 1a is provided in Appendix II. This table illustrates the institutional responses and resulting scores across all channels and dimensions in the context of Bologna University, offering a concrete example of how the tool supports ecosystem mapping and strategic reflection.

Table 1a – Knowledge Valorisation Self-Assessment Tool

Channel	Factor	n	Question
1 - Academia-Industry joint research & mobility	<i>env</i>	1	National policy frameworks effectively support sustained industry–academia co-creation.
		2	Regional policy frameworks effectively support sustained industry–academia co-creation.
		3	IP governance policies are adapted to enable equitable sharing in joint R&I.
		4	Data governance policies are adapted to enable equitable sharing in joint R&I
		5	Are there formal joint research agreements with the industry?
		6	Are research infrastructures co-governed or co-used with industry (e.g. joint labs, testbeds)?
		7	Is regional or national funding used to support these joint research collaborations?
	<i>org</i>	8	Our governance adapts based on lessons from joint research experiences.
		9	IAM systems are integrated into joint project lifecycles from inception to valorisation.
		10	Coordination with industry partners is based on long-term trust and formal mechanisms.
		11	Is there a dedicated unit or office responsible for managing joint research with industry?
		12	Are industry–academia partnerships formally represented in governance or strategic planning?
		13	Are legal/IP staff involved in the early stages of joint project development?
	<i>ind</i>	14	Researchers receive training or mentoring for working with industrial partners.
		15	Incentive systems actively reward intersectoral collaboration.
		16	Academic–industry experiences are shared, systematized and influence new professional roles.
		17	Are researchers formally authorised to lead or co-lead joint projects with industry?
		18	Are industrial PhD or researcher mobility schemes implemented at the institution?
		19	Are academic career criteria (tenure, evaluation) inclusive of industrial collaboration roles?

2 - Promoting research-driven spin-offs and start-ups (access to financing)	<i>env</i>	20	Regional innovation policies align with our internal spin-off strategies.
		21	The institution is embedded in a broader entrepreneurial ecosystem (public + private).
		22	Is your institution linked to an incubator, accelerator, or regional innovation hub?
		23	Are public funding instruments (e.g. national PoC grants) used to support spin-offs?
		24	Are societal impact or mission-driven start-ups part of institutional innovation goals?
	<i>org</i>	25	Lessons from past start-up experiences (successes and failures) inform strategy.
		26	IAM systems are consistently applied throughout the spin-off lifecycle.
		27	Start-up performance is tracked and used in institutional decision-making.
		28	Is there a formal unit overseeing spin-off procedures (equity, IP, legal)?
		29	Are internal procedures in place for handling researcher-founded ventures?
		30	Are there structured templates or model agreements for IP and equity ownership?
	<i>ind</i>	31	Researchers are supported with structured training (e.g. KEI-type programs).
		32	Entrepreneurial activity is recognised in evaluation or career development.
		33	Staff have access to mentors, advisors, or alumni networks for venture support.
		34	Are researchers eligible to found spin-offs or take entrepreneurial leave?
		35	Do staff from SSH or humanities fields receive support for entrepreneurial pathways?
		36	Do researchers receive training in entrepreneurship or start-up creation?
3 - Intermediaries and knowledge transfer professionals	<i>env</i>	37	External networks and policy actors view our KT professionals as strategic collaborators.
		38	Our KT staff participate in co-creation spaces with industry, government, and society.
		39	Does your university participate in national/regional KT networks?
		40	Are liaison offices or coordination platforms (e.g. stakeholder databases) operational?
		41	Do intermediaries regularly contribute to strategy or policy engagement?
	<i>org</i>	42	Intermediary experience is captured and reused in improving institutional policy.
		43	IAM tools are accessible and standardised across departments.
		44	Intermediaries collaborate fluidly across research, engagement, and entrepreneurship.
		45	Are intermediary roles (KT officers, data stewards) formally defined in HR or structures?
		46	Are there cross-unit protocols for contract negotiation, licensing, and brokerage?
		47	Are tools in place to support lifecycle monitoring of KT activities?
	<i>ind</i>	48	Intermediaries receive structured, continuous training.
		49	Feedback mechanisms (e.g. debriefs, evaluations) inform KT practice.
		50	Knowledge brokers operate in empowered, well-resourced roles.
		51	Are intermediary staff part of formal hiring and evaluation structures?
		52	Do staff have access to institutional career progression tied to KT performance?
		53	Are intermediaries represented in strategic institutional discussions?
4 - Engagement of citizens, public bodies	<i>env</i>	54	External societal actors are consistently involved in setting research agendas.
		55	Ethical, inclusive, and transparent data governance is ensured in engagement projects.

and societal actors	<i>org</i>	56	Are there formal frameworks or agreements in place for engagement with societal actors (e.g. schools, NGOs, municipalities)?
		57	Are IAM systems adapted to address data sharing and ethics in participatory research?
		58	Are engagement infrastructures (e.g. living labs, observatories) co-developed with external partners?
		59	Engagement outcomes are integrated into strategic planning and institutional reporting.
		60	Feedback from civil society is systematically gathered and used for improvement.
		61	There are internal reflexive learning processes (e.g. debriefs, evaluations) based on societal interaction.
		62	Is there a designated office, unit, or team responsible for public engagement?
		63	Is citizen engagement formally included in institutional strategy or mission statements?
		64	Are long-term engagement platforms (e.g. civic labs, storytelling labs) strategically resourced?
	<i>ind</i>	65	Staff have regular access to training in science communication and participatory design.
		66	Researchers are encouraged and supported to experiment with novel engagement formats.
		67	Engagement is viewed and rewarded as a legitimate dimension of research and teaching.
		68	Are researchers or staff trained in participatory methods, stakeholder dialogue, or public-facing science?
		69	Are citizen engagement activities considered in performance reviews or internal awards?
		70	Are engagement roles explicitly included in job descriptions or academic career tracks?
5 - Intellectual Property management & Standardisation	<i>env</i>	71	Institutional IP policies are consistently aligned with regional/national legal frameworks.
		72	The institution is proactive in contributing to emerging European or global standards.
		73	Are there clear internal procedures for IP registration and protection?
		74	Is your institution actively involved in national or international standardisation bodies?
		75	Are institutional policies aligned with the EU Code of Practice on Standardisation and IP Management?
	<i>org</i>	76	IP and standards data (e.g. patents, licensing revenue, open tech) are used in decision-making.
		77	IAM systems are interoperable and harmonised across departments.
		78	Internal IP or standardisation failures are used as learning tools to improve governance.
		79	Is there a unit responsible for IP strategy and standardisation support (e.g. IP office, legal desk)?
		80	Are institutional policies for IP integrated with open science, licensing, and innovation goals?
		81	Are licensing and IP use tracked and linked to impact reporting?
	<i>ind</i>	82	Staff are actively incentivised to engage in IP and standardisation pathways.
		83	Researchers understand and apply strategies for balancing open and proprietary outputs.
		84	Professional development includes exposure to IP-related regulatory and business models.

6 - Knowledge circulation & informing policy		85	Are patent authorship, licensing contributions, and standardisation activities reflected in researcher evaluations?
		86	Are training opportunities on IP, licensing, and standards formally offered to researchers?
		87	Are researchers supported in drafting or contributing to technical standards?
	<i>env</i>	88	The institution actively contributes to shaping regional or national research and innovation agendas.
		89	Public authorities engage in ongoing dialogue with institutional researchers.
		90	Are there established mechanisms for sharing policy-relevant knowledge (e.g. policy briefs, evidence hubs)?
		91	Are knowledge-sharing platforms (e.g. FAIRy Day, white paper portals) co-designed with policy actors?
		92	Are HEI outputs routinely used by local, national, or EU-level policy institutions?
	<i>org</i>	93	Institutional learning is informed by policy feedback loops (e.g. consultations, legislative outcomes).
		94	Internal knowledge-sharing processes are adapted to respond to external policy demands.
		95	IAM and Open Science infrastructures are aligned with policy use and traceability standards.
		96	Are there units or processes tasked with translating research into policy-oriented formats?
		97	Are strategic partnerships with public authorities or ministries actively maintained?
		98	Are academia-policy makers relations systematically tracked within the HEI?
	<i>ind</i>	99	Researchers are supported to develop outputs tailored to non-academic audiences.
		100	Contributions to public value and evidence-based policy are encouraged and rewarded.
		101	Skills for societal impact and policy engagement are integrated into doctoral and staff training.
		102	Are contributions to policymaking (e.g. expert panels, regulatory input) recognised in evaluations?
		103	Are researchers trained in strategic foresight, evidence synthesis, policy communication?
		104	Do researchers receive support to participate in policy forums, consultations, advisory boards?

ANNEX II – The case study of Bologna

This annex presents a structured overview of the University of Bologna's knowledge valorisation ecosystem, developed as a pilot case within the KALEIDOS project.

The first step in the process involved a detailed mapping exercise to visualise how key actors—across academia, industry, government, and civil society—interact within each of the six EU-defined valorisation channels. By analysing the environmental, organisational, and individual dimensions for each channel, this initial reflection provided a situated example of how valorisation practices are configured, resourced, and evolving within a mature regional innovation context.

Building on this foundational mapping, the second step was the completion of the KALEIDOS self-assessment tool. This was carried out by the ARIN – Knowledge Transfer Office – New Entrepreneurship Unit, which coordinates institutional activities related to innovation, IP, and stakeholder engagement. The self-assessment generated a composite picture of institutional readiness based on scores from both closed (Yes/No) and scaled (1–7) diagnostic questions. Each factor within a channel—environmental, organisational, or individual—produced a partial score, which contributed to the channel-level score. These six channel scores were then used to construct an overall institutional maturity profile. The full set of diagnostic questions, presented in Annex I, guided the assessment process.

The resulting channel-level readiness scores ranged from 4.9 to 6.2, with an overall institutional average of 5.6 out of 7, placing UniBo within the high readiness band according to KALEIDOS scoring thresholds (Low = 1.0–3.4; Medium = 3.5–5.4; High = 5.5–7.0). UniBo demonstrated strong performance in areas such as Spin-offs and Start-ups (6.2), Joint Research with Industry (5.9), and IP Management (5.6). More variable results emerged in Knowledge Circulation and Policy Engagement (4.9), where environmental and organisational integration was less developed. At the factor level, environmental dimensions consistently scored highest, reflecting Bologna's embeddedness in a well-resourced and policy-aligned regional innovation system. Organisational scores showed greater variability, while individual factors—linked to incentives, training, and role clarity—highlighted the need for targeted capacity-building across channels.

Together, the mapping and readiness assessment provide a comprehensive portrait of UniBo's valorisation ecosystem. As illustrated in Table 2a and Figures 1a and 2a, this two-step approach offers a replicable model for other institutions seeking to diagnose their current configuration, identify leverage points, and build more intentional and coordinated knowledge valorisation strategies.

Table 2a – Mapping of Bologna Ecosystem

1. Academia–Industry Joint Research & Mobility			
ENV	Framework agreements with major industrial players (e.g. STMicroelectronics, RFI) anchored in regional and EU R&I ecosystems (e.g. ECOSISTER, S3 strategy). These provide long-term vision and policy alignment for co-creation.	Joint research labs (e.g. ARCES–STMicroelectronics, ARCES–RFI) act as boundary infrastructures, providing a space where academic and industrial researchers collaborate daily, co-developing technologies and sharing equipment.	The relationship between actors is reciprocal but asymmetric: while industry seeks applicability and speed, academia prioritises research depth and publishing.
ORG	Joint labs hosted by ARCES, supported by infrastructure like Pepoli Lab and the Cesena Campus. Co-governance models coordinate access to spaces, data, and human resources. Also includes CIRI and PhD with Industry programs.	PhD with industry programs and CIRI initiatives mediate long-term joint agendas and embed students in industrial contexts.	A long feedback loop connects lab research → prototyping → industrial application → new research questions → renewed collaborations.
IND	Researchers from academia and industry work side-by-side, co-supervise PhDs, and engage in applied R&I. Staff mobility and mutual upskilling are part of the culture, with a strong emphasis on interdisciplinary project experience.	Framework agreements and shared governance documents facilitate coordination of IP, access to data, and research goals.	These dynamics promote co-evolution of expertise, infrastructure, and culture. Institutional trust builds over time, enabling more fluid mobility and shared risk.
QH ACTORS	Academia ARCES, CIRI, departments Industry STMicroelectronics, RFI, Fraunhofer Government Ministry of University and Research (MUR), Emilia-Romagna Region Civil Society Indirectly via downstream applications (mobility, energy)		
2. Spin-offs, Start-ups & Access to Finance			
ENV	The channel is embedded in the Emilia-Romagna innovation ecosystem. Supported by PR FESR, NextGenerationEU, and platforms like EROI and initiatives like Startup Day. Linked to the AlmaCube incubator and digital transformation initiatives.	ALMALABOR and I.D.E.A. function as knowledge-to-business accelerators, translating academic ideas into market-ready prototypes.	This is a pipeline-oriented system, with distinct phases of ideation, prototyping, validation, and spin-off creation.
ORG	The channel is operated through structures like ALMALABOR, the KEI Master, and I.D.E.A.. These create a pipeline from idea generation to proof-of-concept, with access to mentors, prototyping facilities, and funding schemes.	Pitching events, incubators, and spin-off calls are structured rituals that convert informal interest into structured entrepreneurial journeys.	The dynamics involve constant negotiation between risk and vision: is an idea ready for market? Who owns what? Is societal value being diluted?
IND	Students, PhD candidates, and postdocs are offered training in pitching, IP, business development, and venture creation. Coaching and mentoring networks support transition from academic output to entrepreneurial projects.	KEI Master and mentoring programs serve as educational and social infrastructures where researchers learn to speak the language of investors and IP officers.	There's a circular flow: spin-off founders return as mentors, evaluators, or collaborators, reinforcing the ecosystem.
QH ACTORS	Academia ALMALABOR, I.D.E.A., KEI program Industry AlmaCube, private investors, accelerators Government regional innovation funds, ERDF Civil Society social entrepreneurs, cooperative innovators		Start-up success (or failure) feeds back into the university's policies on IP, PoC funding, and entrepreneurial education.

3. Intermediaries & Knowledge Transfer Professionals			
ENV	Participation in Netval, ART-ER, and international KT networks. Legal infrastructure supports contracting, co-ownership, and shared risk management. The model aligns with EU and national KT policy frameworks.	Knowledge Transfer Office (KTO) and departmental KT Juniors act as internal brokers, mediating between researchers, legal units, and external partners.	This system is shaped by constant flow across cultural logics: academic curiosity, legal compliance, commercial ambition.
ORG	UniBo operates a hybrid KTO model: a central office plus KT Juniors embedded in departments. This facilitates decentralised interaction while maintaining alignment through training, IP support, and data tracking.	Standard contracts, IP checklists, and valuation matrices serve as boundary objects that standardise communication across disciplines and sectors.	The KTO balances responsiveness and risk control: enabling quick collaboration while safeguarding ownership and ethical boundaries.
IND	KT Junior Managers and staff are upskilled in contract law, IP licensing, EU valorisation tools, and brokerage skills. Their professionalisation is central to UniBo's knowledge strategy and aligns with EU Codes of Practice.	External partners (e.g. ART-ER, Netval) function as meta-intermediaries, connecting UniBo with national and European networks.	While informal learning occurs across collaborations, the absence of dedicated knowledge brokers or structured feedback mechanisms limits systematic adaptation. Process revisions—when they happen—tend to be reactive and decentralised, rather than part of an institutional learning framework..
QH ACTORS	Academia KTO, KT Juniors, Data Stewards Industry client companies, legal IP consultants Government regional innovation funds, ERDF Civil Society ART-ER, Netval, regional tech clusters		
4. Engagement of Citizens, Public Bodies & Societal Actors			
ENV	Citizen engagement is formally embedded in UniBo's Strategic Plan 2022–2027 (Goal 41). Aligned with the EU Code of Practice on Citizen Engagement and regional public participation frameworks.	REAL Lab, Open BOOK, and UniBo Magazine act as engagement platforms that allow non-academic actors to shape, access, or comment on research.	Engagement is not linear: it fluctuates with relevance, accessibility, and trust.
ORG	Structures such as the REAL Lab, Open BOOK, and storytelling labs serve as institutional vehicles for participatory science, societal dialogue, and co-creation with citizens and schools.	Events like PhD Storytelling Labs, co-design workshops, and public consultations provide structured formats for interaction and influence.	A key dynamic is development through storytelling: researchers must learn to reframe complex problems in ways that resonate with lay audiences.
IND	Researchers and students are trained in science communication, narrative methods, and co-design. PhD labs emphasise storytelling, while community events offer real-world participatory experience.	Citizen science tools, such as feedback forms, participatory foresight templates, or hackathon kits, transform informal contributions into recognised inputs.	Tensions often emerge between depth of engagement vs institutional timelines: co-design takes time, yet funding and project cycles are short.
QH ACTORS	Academia REAL Lab, Open BOOK, storytelling labs Industry partners in co-design (e.g., in CityLabs) Government municipal + regional civic innovation initiatives Civil Society teachers, students, citizens		Positive feedback loops occur when citizen insights reshape research agendas or when engagement fosters new funding (e.g. societal challenge-oriented grants).
5. Intellectual Property Management & Standardisation			
ENV	UniBo's IP system is aligned with Horizon IP policy, CoARA, and the EU Code of Practice on Standardisation. Frameworks support joint ownership, licensing, and strategic patenting.	IP Managers, KEI-trained staff, and legal consultants facilitate the conversion of research into IP assets.	The dynamic is one of codification: turning tacit or experimental knowledge into discrete, protectable units.
ORG	The IP Office collaborates with the KEI Master, CIRI legal staff, and spin-off advisory units. IP and standardisation processes are integrated into project planning, PoC evaluations, and research contracts.	Joint ownership agreements, standardisation roadmaps, and PoC templates are tools that structure negotiation and co-responsibility.	Tensions arise between openness (e.g., Open Science) and protection (e.g., licensing exclusivity).
IND	Staff and students receive training in IP management, standardisation (ISO, CEN), and licensing models. Joint training formats (e.g. KEI) build capacity for co-creation and responsible innovation.	Co-authored patents and CEN/CENELEC interactions function as integration points between academia and industry.	A reinforcing feedback loop exists when successful IP cases generate revenue or visibility, leading to more interest and participation in IP training and co-creation.
QH ACTORS	Academia KEI-trained staff, IP offices, CIRI legal staff Industry patent co-owners (e.g., ST), tech accelerators Government EU IPR frameworks, CoARA reforms Civil Society open licensing communities		Ethical dilemmas—such as who should benefit from IP developed with public funding—can trigger policy debates and strategy revisions.
6. Knowledge Circulation & Informing Policy			
ENV	Strong alignment with Open Science mandates, FAIR principles, FAIRy Day, CoARA, and JRC policy uptake strategies. Participation in ERA and national open access coalitions.	Open repositories (AMS Acta, IRIS-IR), FAIRy Day, and policy briefs act as formats for structured knowledge dissemination.	This is a complex, multilevel circulation system, where outputs must travel across formats, sectors, and epistemic cultures.
ORG	Open Science is coordinated through GLOS, with Data Stewards in each department. Institutional repositories (AMS Acta, IRIS-IR) enable visibility and traceability of research outputs and datasets.	Data Stewards and GLOS coordinate the flow of data and ensure alignment with international standards (e.g. FAIR, EOSC).	The system thrives on reusability: research data is used by other scholars, policymakers, or societal actors in different contexts.
IND	Researchers are trained in data governance, policy engagement, and public dissemination. Staff are supported in crafting policy briefs, maintaining open repositories, and tracking societal uptake.	Science communication channels (e.g. UniBo social media, video explainers) are increasingly used to reach policymakers and the public.	A core tension is between compliance and genuine openness: publishing in open repositories is not the same as being understood or impactful.
QH ACTORS	Academia data stewards, FAIRy Day organisers Industry advisory boards, foresight partners Government CoARA, JRC, ERA policy designers Civil Society science communicators, NGOs		Dynamic alignment with policy evolves through evidence feedback loops: research informs policy; policy shapes research incentives.

Figure 1a – University of Bologna, Total KV Readiness score by channel

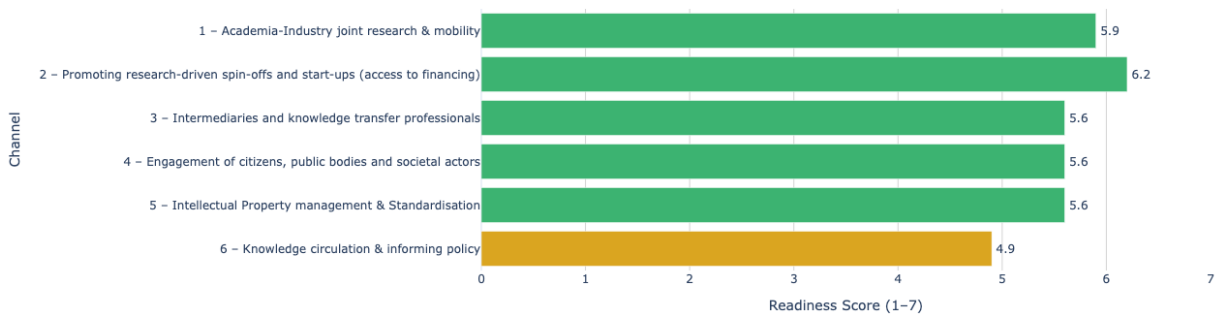


Figure 2a – University of Bologna, Partial Readiness Scores by Factor (ENV / ORG / IND)

